

CHELYABINSK GOROD, Russian Federation

WMO: 286450

Lat: 55.150N

Lon: 61.300E

Elev: 237

StdP: 98.51

Time Zone: 5.00 (E05)

Period: 04-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-27.5	-33.5	0.2	-30.5	-30.0	0.2	-27.1	6.0	-14.8	5.1	-10.3	0.7	0	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	30.4	19.1	28.6	18.9	27.0	18.1	21.0	27.7	20.2	26.3	19.3	25.0	1.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.0	24.0	17.9	13.2	23.4	17.1	12.5	22.4	61.9	27.8	59.1	26.4	55.9	25.2	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.8	4.1	DB	-34.7	33.2	4.3	3.1	-37.8	35.4	-40.4	37.2	-42.8	38.9	-45.9	41.2	(4)
(5)			WB	-34.8	22.5	4.3	0.7	-37.9	22.9	-40.4	23.3	-42.9	23.7	-46.0	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-15.7	-13.4	-5.3	5.5	13.4	18.2	19.4	18.0	11.9	4.9	-4.0	-11.6	(6)	
	DBStd	13.55	6.45	6.02	4.93	5.02	4.71	4.36	4.06	3.99	4.26	4.54	6.64	7.21	(7)	
	HDD10.0	3388	798	657	474	149	21	3	0	1	27	167	419	671	(8)	
	HDD18.3	5588	1056	890	733	384	166	54	37	56	198	416	669	929	(9)	
	CDD10.0	1023	0	0	0	15	126	248	293	248	83	9	0	0	(10)	
	CDD18.3	181	0	0	0	0	11	49	72	46	4	0	0	0	(11)	
	CDH23.3	1599	0	0	0	7	146	440	614	360	32	0	0	0	(12)	
	CDH26.7	425	0	0	0	1	23	116	179	103	4	0	0	0	(13)	
(14) Wind	WSAvg	1.6	1.3	1.4	1.9	2.0	1.7	1.5	1.4	1.3	1.4	1.7	1.8	1.4	(14)	
(15) Precipitation	PrecAvg	645	43	32	40	38	53	73	87	70	50	58	54	46	(15)	
	PrecMax	820	107	64	86	100	115	123	191	116	94	131	101	93	(16)	
	PrecMin	462	9	8	3	3	15	35	37	22	7	9	5	14	(17)	
	PrecStd	93	21	14	21	23	26	22	40	26	20	29	24	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.4	10.1	24.0	29.2	31.9	33.2	32.5	27.3	18.7	9.0	4.5	(19)	
		MCWB	0.2	1.4	5.4	13.4	16.3	19.1	19.9	19.5	16.6	11.2	6.0	1.5	(20)	
	2%	DB	-1.4	0.5	5.6	19.5	27.0	29.8	30.3	29.3	23.6	16.5	6.6	1.9	(21)	
		MCWB	-2.7	-1.2	2.8	10.7	15.5	18.4	19.6	19.3	15.5	10.2	4.5	0.4	(22)	
	5%	DB	-3.7	-1.7	3.4	16.3	25.0	27.6	28.6	27.1	21.1	14.2	4.9	0.4	(23)	
		MCWB	-4.7	-3.4	1.1	9.1	15.1	18.4	19.6	19.0	14.6	9.2	3.3	-0.7	(24)	
	10%	DB	-5.8	-4.3	1.8	13.5	22.5	25.9	26.8	24.9	18.6	11.8	3.4	-1.4	(25)	
		MCWB	-6.7	-5.7	0.0	7.8	13.9	17.7	18.7	18.1	13.5	8.2	1.9	-2.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	1.6	6.2	13.5	18.4	21.4	22.1	21.4	17.6	12.1	6.1	1.8	(27)	
		MCDB	1.9	3.3	9.7	23.0	25.0	28.7	28.2	28.3	24.6	17.1	8.4	3.9	(28)	
	2%	WB	-2.4	-1.0	3.2	11.2	16.7	20.2	21.3	20.6	16.3	10.8	4.6	0.4	(29)	
		MCDB	-1.6	0.3	5.6	17.7	24.4	26.9	27.7	27.3	22.1	15.2	6.4	1.8	(30)	
	5%	WB	-4.6	-3.4	1.6	9.7	15.7	19.1	20.6	19.7	15.2	9.6	3.5	-0.7	(31)	
		MCDB	-3.7	-1.8	3.0	15.4	23.4	25.5	26.6	25.7	20.4	13.3	4.8	0.4	(32)	
	10%	WB	-6.7	-5.5	0.2	8.2	14.6	18.2	19.7	18.7	14.0	8.4	2.0	-2.6	(33)	
		MCDB	-5.8	-4.3	1.6	13.2	21.8	24.5	25.0	24.1	17.6	11.6	3.3	-1.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	11.2	9.7	10.0	11.4	10.1	9.6	9.1	8.8	7.4	5.6	7.0	(35)	
		MCDBR	7.5	10.5	9.9	14.8	14.6	13.1	12.7	12.3	12.1	11.4	6.2	4.9	(36)	
	5% WB	MCWBR	6.7	9.0	7.3	7.6	5.9	5.5	4.6	5.0	5.5	6.3	4.7	3.9	(37)	
		MCWBR	7.0	9.8	9.0	14.0	13.3	11.5	10.7	11.3	10.8	10.7	6.1	5.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.282	0.311	0.367	0.408	0.408	0.405	0.432	0.445	0.385	0.354	0.309	0.280	(40)	
		taud	2.073	2.046	1.978	2.123	2.230	2.310	2.223	2.161	2.293	2.255	2.165	2.048	(41)	
	Ebn at Noon		614	737	775	807	834	842	806	764	767	690	579	514	(42)	
		Edh at Noon	66	98	135	135	129	121	129	129	99	81	61	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.87	1.94	3.44	4.41	5.23	5.68	5.51	4.31	2.93	1.49	0.75	0.54	(44)	
	RadStd		0.08	0.11	0.24	0.45	0.50	0.50	0.39	0.42	0.23	0.26	0.09	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page